QML Lab • research project 04

Re-uploading Depth Study on MNIST

How do layers and qubits trade off in a small data re-uploading classifier for handwritten digits?

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Quadrant	CQ (classical data, quantum processor)	**Track**	Variational
Difficulty	Easy	**Tools**	PennyLane, scikit-learn
Builds on	Module 5 of the QML Lab	**Workspace**	`qml_projects/reuploading-depth-mnist/`

Summary. A data re-uploading circuit encodes the input again before every trainable layer, which lets even a few qubits express complicated functions. This project maps how test accuracy on pairs of handwritten digits depends on the number of layers and qubits, and compares every configuration with logistic regression and a small MLP on the same PCA features. It is the entry project of the lab and already has a runnable experiment.

1 Background and key concepts

1.1 Digits and PCA

MNIST (LeCun et al., 1998) has 28×28 images of handwritten digits. The scikit-learn digits set, used here first, is the smaller UCI collection of 8×8 images (Alpaydin and Kaynak, 1998), with about 180 images per class. PCA reduces each image to n components, one per qubit, scaled to $[0, \pi]$.

1.2 Variational classifier

As in Module 5, the model is the expectation value of Z on the first qubit after a parameterised circuit,

$$f_\theta(x) = \langle 0 | \, U(x,\theta)^\dagger Z_0 \, U(x,\theta) \, |0\rangle + b, \qquad U(x,\theta) = \prod_{\ell=1}^{L} W(\theta_\ell)\, S(x), \tag{1}$$

where $S(x)$ is angle encoding and $W(\theta_\ell)$ a strongly entangling layer with $3n$ angles. The classifier has $3nL + 1$ trainable parameters (Schuld et al., 2020). Training minimises the mean squared error to labels ± 1 with Adam (Kingma and Ba, 2015).

1.3 Why re-uploading helps

Schuld et al. (2021) showed that such a model is a truncated Fourier series in the inputs,

$$f_\theta(x) = \sum_{\omega \in \Omega} c_\omega(\theta)\, e^{i\omega \cdot x}, \tag{2}$$

and that each repetition of the encoding adds frequencies to Ω. With one encoding the model is a single sine per feature; with L repetitions it reaches frequency L (Pérez-Salinas et al., 2020). Layers therefore buy expressivity, while qubits buy input features.

2 Research questions

The project is open-ended: these questions are the starting point, not the finish line. Add new ones to the README and the journal as they come up.

RQ1 Where does adding layers stop improving test accuracy on 4 PCA features?

RQ2 For a fixed parameter budget, is it better to add qubits (features) or layers (depth)?

RQ3 How far is the best circuit from logistic regression and a small MLP on the same features?

3 Method

3.1 Data

Digit pairs from the scikit-learn digits set: 3 vs 5 (default), 4 vs 9 and 1 vs 7. A stratified 70/30 train/test split per seed; PCA and scaling are fitted on the training part only. Later, full MNIST reduced with PCA.

3.2 Models

The existing `baseline` experiment: n qubits, L re-uploading layers of `AngleEmbedding` plus `StronglyEntanglingLayers`, MSE loss, Adam with learning rate 0.1, mini-batches of 32, 15 epochs.

3.3 Baselines

Logistic regression on the same PCA features (already logged as `baseline_accuracy`); an MLP and an RBF-SVM with matched parameter counts.

3.4 Metrics

- Test accuracy (mean $\pm$ s.d. over 5 seeds) and final training loss.
- Number of trainable weights; training time per epoch.

4 Planned experiments

Each experiment becomes one `[[experiments]]` entry in the workspace, so its runs, parameters and metrics are stored and compared in the portal.

5 Running it in the QML Lab

The workspace lives in `qml_projects/reuploading-depth-mnist/`. Scripts import three helpers from `qml_lab/plots.py`:

- `params()` reads the parameters from the portal's run form (defaults when run by hand);
- `save()` stores figures with the run;

ID	Experiment	What varies	Main metrics
E1	Depth sweep	$L \in \{1,2,4,6\}$, $n = 4$	test accuracy
E2	Qubit sweep	$n \in \{2,4,6,8\}$, $L = 2$	test accuracy
E3	Equal budget	(n, L) pairs with $\approx$ same $3nL$	accuracy per parameter
E4	Hard pairs	3 vs 5, 4 vs 9, 1 vs 7	gap to baselines
E5	Full MNIST	28×28 images, PCA	accuracy, run time

Table 1: Experiment plan. Results go into the journal and the results log (section 9).

- `log_metrics()` records numbers that appear in the run table.

```
qml_projects/reuploading-depth-mnist/
  project.toml          metadata + experiments
  README.md             goal, open questions, ideas
  experiments/          runnable scripts
  docs/main.tex         this document
```

First experiment: This experiment already exists in the workspace and can be run from the project page.

```
[[experiments]]
slug = "baseline"
title = "Re-uploading classifier vs logistic regression"
script = "experiments/e01_baseline.py"
figure = "baseline.png"

[experiments.params]
digit_a = 3
digit_b = 5
n_qubits = 4
n_layers = 2
epochs = 15
seed = 0
```

```
# from the portal folder: pick up changes to project.toml
python manage.py sync_projects
cd qml_projects/reuploading-depth-mnist
# run by hand with the default parameters
PYTHONPATH=../../qml_lab python experiments/e01_baseline.py
```

6 Quality bar

- A classical baseline with a similar number of trainable parameters, tuned with the same care (Bowles et al., 2024).
- At least five random seeds per configuration, reported as mean $\pm$ standard deviation.
- How results change with the number of qubits.
- At least one run under a realistic noise model.
- Fit PCA and scaling on training data only, so the test set stays unseen.

7 Open directions

- Multi-class classification with one readout qubit per class.
- Amplitude encoding of the full 8×8 image on 6 qubits.
- Trainable input scaling, which changes the available Fourier frequencies.
- Repeat the best configuration under a noise model (Project 14).

8 Risks and limitations

- Small test sets (about 100 images per pair) give noisy accuracies; always use several seeds.
- PCA features of easy pairs are almost linearly separable, so all models may hit a ceiling.

9 Results log

Summarise findings here as they become stable; day-to-day notes belong in the portal journal.

Date	Run	Finding	Next step

References

Alpaydin, E. and Kaynak, C. (1998) 'Optical recognition of handwritten digits', UCI Machine Learning Repository. Available at: https://doi.org/10.24432/C50P49.

Bowles, J., Ahmed, S. and Schuld, M. (2024) 'Better than classical? The subtle art of benchmarking quantum machine learning models', Preprint, arXiv:2403.07059. Available at: https://arxiv.org/abs/2403.07059.

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LeCun, Y., Bottou, L., Bengio, Y. and Haffner, P. (1998) 'Gradient-based learning applied to document recognition', *Proceedings of the IEEE*, 86(11), pp. 2278–2324. Available at: https://doi.org/10.1109/5.726791.

Pérez-Salinas, A., Cervera-Lierta, A., Gil-Fuster, E. and Latorre, J. I. (2020) 'Data re-uploading for a universal quantum classifier', *Quantum*, 4, 226. Available at: https://doi.org/10.22331/q-2020-02-06-226.

Schuld, M., Bocharov, A., Svore, K. M. and Wiebe, N. (2020) 'Circuit-centric quantum classifiers', *Physical Review A*, 101(3), 032308. Available at: https://doi.org/10.1103/PhysRevA.101.032308.

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